The Geometry of a Perceptron

1 Summary under normalisation assumptions

Consider the perceptron

$$s(\mathbf{x}) = \mathbf{w}^\mathsf{T}\mathbf{x} + b, \qquad \widehat{y} = \mathbf{1}_{\{s(\mathbf{x})\geq 0\}}.$$

Assume $\mathbf{w} \neq \mathbf{0}$. We may normalise the weights without changing the classifier by replacing

$$\mathbf{w} \longmapsto \frac{\mathbf{w}}{\|\mathbf{w}\|}, \qquad b \longmapsto \frac{b}{\|\mathbf{w}\|}.$$

Both must be rescaled together to preserve the decision boundary. In the following, $\mathbf{w}$ and b denote these rescaled parameters, so $\|\mathbf{w}\| = 1$.

1.1 Normalised weights

With $\|\mathbf{w}\| = 1$,

$$s(\mathbf{x}) = \mathbf{w}^\mathsf{T}\mathbf{x} + b$$

has two geometric interpretations:

- $\mathbf{w}^\mathsf{T}\mathbf{x}$ is the scalar projection of the input onto the weight direction.
- $s(\mathbf{x})$ is the signed perpendicular distance from the input to the decision hyperplane.

The boundary and classification rule are

$$\mathcal{H} = \{\mathbf{x} : \mathbf{w}^\mathsf{T}\mathbf{x} = -b\}, \qquad \widehat{y} = 1 \iff \mathbf{w}^\mathsf{T}\mathbf{x} \geq -b.$$

Thus $\mathbf{w}$ determines the orientation, and $-b$ determines the threshold along the normal direction.

1.2 Normalised weights and inputs

If we also assume $\|\mathbf{x}\| = 1$, then

$$\mathbf{w}^\mathsf{T}\mathbf{x} = \cos\alpha, \qquad s(\mathbf{x}) = \cos\alpha + b,$$

where $\alpha \in [0, \pi]$ is the angle between the input and weight vectors. Therefore,

$$\widehat{y} = 1 \iff \cos\alpha \geq -b.$$

For $-1 < b < 1$, this becomes

$$\widehat{y} = 1 \iff \alpha \leq \arccos(-b).$$

Class 1 is therefore a spherical cap centred on the weight direction. When $b = 0$, this cap is a hemisphere:

$$\widehat{y} = 1 \iff \alpha \leq \frac{\pi}{2}.$$

The limiting cases are:

- If $b \geq 1$, every unit input receives class 1.
- If $b < -1$, every unit input receives class 0.
- If $b = -1$, only $\mathbf{x} = \mathbf{w}$ receives class 1.

Normalising each input is a modelling choice: it removes information about its magnitude and can change the classification problem. Unlike jointly rescaling the weights and bias, it does not generally preserve the original classifier.

1.3 Regularity and sensitivity

The score is an affine, continuously differentiable function, with

$$\nabla_{\mathbf{x}} s = \mathbf{w}.$$

For unit weights, the Cauchy–Schwarz inequality gives

$$|s(\mathbf{x} + \boldsymbol{\delta}) - s(\mathbf{x})| = |\mathbf{w}^{\mathsf{T}} \boldsymbol{\delta}| \leq \|\boldsymbol{\delta}\|.$$

Consequently, if $s(\mathbf{x}) \neq 0$, any perturbation satisfying

$$\|\boldsymbol{\delta}\| < |s(\mathbf{x})|$$

preserves the predicted class.

The score is smooth, but the thresholded prediction is discontinuous at the decision boundary. Normalisation does not remove this discontinuity.

2 A perceptron in two dimensions

Represent an input as

$$\mathbf{x} = (x_1, x_2),$$

and choose a unit weight vector pointing in the direction θ:

$$\mathbf{w} = (\cos\theta, \sin\theta).$$

The perceptron computes the score

$$s = \mathbf{w}^{\mathsf{T}} \mathbf{x} + b = x_1 \cos\theta + x_2 \sin\theta + b,$$

where $b \in \mathbb{R}$ is the bias. It then applies the classification rule

$$\widehat{y} = \begin{cases} 1, & s \geq 0, \\ 0, & s < 0. \end{cases}$$

Because $\|\mathbf{w}\| = 1$, the dot product $\mathbf{w}^{\mathsf{T}} \mathbf{x}$ is the scalar projection of the input onto the weight direction. The bias determines the threshold along that direction:

$$\widehat{y} = 1 \quad \Longleftrightarrow \quad \mathbf{w}^{\mathsf{T}} \mathbf{x} \geq -b.$$

2.1 The decision boundary

The decision boundary consists of points satisfying

$$\mathbf{w}^\mathsf{T}\mathbf{x} + b = 0,$$

or equivalently,

$$x_1 \cos\theta + x_2 \sin\theta + b = 0.$$

This is a line perpendicular to the weight vector. The weights determine its normal direction, while the bias determines its position. When $b = 0$, the line passes through the origin.

For example, let $\theta = \pi/4$ and $b = 0$. Then

$$\mathbf{w} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \qquad s = \frac{x_1 + x_2}{\sqrt{2}}.$$

The perceptron predicts class 1 when $x_1 + x_2 \geq 0$. Its decision boundary is

$$x_2 = -x_1.$$

2.2 The angular interpretation

If the input is also normalised, we can write

$$\mathbf{x} = (\cos\phi, \sin\phi).$$

The score becomes

$$\begin{aligned} s &= \cos\theta\cos\phi + \sin\theta\sin\phi + b \\ &= \cos(\phi - \theta) + b. \end{aligned}$$

With $b = 0$, the perceptron predicts class 1 whenever the angle between the input and weight directions is at most $90°$. Inputs at exactly $90°$ lie on the boundary and are assigned class 1 under our convention.

3 Extension to higher dimensions

For an input with n features, let

$$\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n, \qquad \mathbf{w} = (w_1, \ldots, w_n) \in \mathbb{R}^n.$$

The score and classification rule remain

$$s = \mathbf{w}^\mathsf{T}\mathbf{x} + b = \sum_{i=1}^{n} w_i x_i + b,$$

$$\widehat{y} = \begin{cases} 1, & s \geq 0, \\ 0, & s < 0. \end{cases}$$

Each weight w_i determines the contribution of the corresponding feature x_i to the score. Together, the weights define a direction in the feature space.

Throughout the geometric discussion, assume $\mathbf{w} \neq \mathbf{0}$. If $\mathbf{w} = \mathbf{0}$, the score is the constant b, so the perceptron does not distinguish between inputs.

3.1 Projection onto the weight direction

Define the unit weight direction

$$\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|}.$$

Then

$$s = \|\mathbf{w}\|\,\mathbf{u}^{\mathsf{T}}\mathbf{x} + b.$$

The quantity $\mathbf{u}^{\mathsf{T}}\mathbf{x}$ is the scalar projection of $\mathbf{x}$ onto the weight direction. Therefore,

$$\widehat{y} = 1 \quad \Longleftrightarrow \quad \mathbf{u}^{\mathsf{T}}\mathbf{x} \geq -\frac{b}{\|\mathbf{w}\|}.$$

Although the input has n coordinates, a single perceptron reduces them to one scalar projection and compares that projection with a threshold.

3.2 The decision hyperplane

The decision boundary is

$$\mathcal{H} = \left\{\mathbf{x} \in \mathbb{R}^n : \mathbf{w}^{\mathsf{T}}\mathbf{x} + b = 0\right\}.$$

For nonzero $\mathbf{w}$, this is an $(n-1)$-dimensional affine hyperplane.

Input space	Decision boundary
$\mathbb{R}^2$	A line
$\mathbb{R}^3$	A plane
$\mathbb{R}^n$	An $(n-1)$-dimensional hyperplane

To verify that $\mathbf{w}$ is normal to the boundary, take any two points $\mathbf{p}, \mathbf{q} \in \mathcal{H}$. Then

$$\mathbf{w}^{\mathsf{T}}\mathbf{p} + b = 0, \qquad \mathbf{w}^{\mathsf{T}}\mathbf{q} + b = 0.$$

Subtracting gives

$$\mathbf{w}^{\mathsf{T}}(\mathbf{p} - \mathbf{q}) = 0.$$

Thus every displacement along the hyperplane is orthogonal to the weight vector.

3.3 A three-dimensional example

Choose

$$\mathbf{w} = \frac{1}{\sqrt{3}}(1, 1, 1), \qquad b = -1.$$

The score is

$$s = \frac{x_1 + x_2 + x_3}{\sqrt{3}} - 1,$$

and the decision boundary is the plane

$$x_1 + x_2 + x_3 = \sqrt{3}.$$

The perceptron predicts class 1 when

$$x_1 + x_2 + x_3 \geq \sqrt{3}.$$

The weight direction has equal components along the three coordinate axes. Moving in that direction increases the score, while moving parallel to the plane leaves the score unchanged.

More generally, for any displacement $\mathbf{v}$,

$$s(\mathbf{x} + \mathbf{v}) - s(\mathbf{x}) = \mathbf{w}^\mathsf{T}\mathbf{v}.$$

Consequently, a displacement orthogonal to $\mathbf{w}$ does not change the score.

4 Parallel and perpendicular components

Using the unit weight direction $\mathbf{u}$, decompose the input as

$$\mathbf{x} = \mathbf{x}_\parallel + \mathbf{x}_\perp,$$

where

$$\mathbf{x}_\parallel = (\mathbf{u}^\mathsf{T}\mathbf{x})\mathbf{u}, \qquad \mathbf{x}_\perp = \mathbf{x} - (\mathbf{u}^\mathsf{T}\mathbf{x})\mathbf{u}.$$

Since

$$\mathbf{u}^\mathsf{T}\mathbf{x}_\perp = 0,$$

the score depends only on the component parallel to the weight direction:

$$s = \|\mathbf{w}\|\, \mathbf{u}^\mathsf{T}\mathbf{x}_\parallel + b.$$

All inputs with the same projection onto $\mathbf{u}$ receive the same score, regardless of their perpendicular components.

5 Angles in higher dimensions

For nonzero vectors $\mathbf{w}$ and $\mathbf{x}$, let $\alpha \in [0, \pi]$ be their angle. The dot product satisfies

$$\mathbf{w}^\mathsf{T}\mathbf{x} = \|\mathbf{w}\|\, \|\mathbf{x}\| \cos\alpha.$$

Hence

$$s = \|\mathbf{w}\|\, \|\mathbf{x}\| \cos\alpha + b.$$

If both vectors are normalised,

$$s = \cos\alpha + b.$$

With zero bias, class 1 contains directions making an angle of at most 90° with $\mathbf{w}$. On the unit sphere, the two classes occupy opposing hemispheres, with the shared boundary assigned to class 1.

In two dimensions, a unit direction can be described by one angle. In n dimensions, a unit direction generally requires $n - 1$ angular parameters, but the dot-product formulation avoids the need to introduce them explicitly.

6 Signed distance to the boundary

The signed perpendicular distance from $\mathbf{x}$ to $\mathcal{H}$ is

$$d(\mathbf{x}) = \frac{\mathbf{w}^\mathsf{T}\mathbf{x} + b}{\|\mathbf{w}\|}.$$

The nearest point on the hyperplane is

$$\mathbf{x}_\mathcal{H} = \mathbf{x} - \frac{\mathbf{w}^\mathsf{T}\mathbf{x} + b}{\|\mathbf{w}\|^2}\mathbf{w}.$$

Indeed,

$$\mathbf{w}^\mathsf{T}\mathbf{x}_\mathcal{H} + b = 0,$$

and $\mathbf{x} - \mathbf{x}_\mathcal{H}$ is parallel to the normal vector $\mathbf{w}$. The ordinary distance is $|d(\mathbf{x})|$.

Positive signed distances correspond to the class-1 side, negative distances correspond to the class-0 side, and zero corresponds to the boundary. If $\|\mathbf{w}\| = 1$, the score itself is the signed distance.

7 Training and linear separability

Training adjusts $\mathbf{w}$ and b to separate labelled examples. Geometrically, this changes the orientation and position of the decision hyperplane.

Multiplying both weights and bias by any positive constant c gives

$$(c\mathbf{w})^\mathsf{T}\mathbf{x} + cb = c\big(\mathbf{w}^\mathsf{T}\mathbf{x} + b\big).$$

This changes the magnitude of the score but preserves its sign, the decision boundary, and all classifications. It also preserves the signed distance:

$$\frac{(c\mathbf{w})^\mathsf{T}\mathbf{x} + cb}{\|c\mathbf{w}\|} = \frac{\mathbf{w}^\mathsf{T}\mathbf{x} + b}{\|\mathbf{w}\|}.$$

A single perceptron has a hyperplane as its decision boundary. It can perfectly separate labelled classes only when a suitable linear separator exists in the chosen feature space. Adding dimensions provides more features, but the boundary remains a hyperplane in that space.